INTRODUCTION TO ECONOMICS
MGT404E

Risk Aversion and Insurance

Yale

Review and Today's Plan

Last Class

Finishing up price discrimination:

- Perfect, demographic
- Self-selecting
- Quantity
- Two-part tariffs (logic; no math)

So far, we have always assumed firms and individuals are *risk-neutral*.

Today

- Understanding and measuring risk
- The value of insurance
- Mitigating risk

Today's Class

All about risk and uncertainty:

1. Uncertainty and the idea of a Utility Function

- Uncertain outcomes are a natural idea
- Expected value captures a risk-neutral view of the outcomes that are possible
- Utility captures risk-averse behavior.

2. Implications of the Utility Function

- People differ in utility functions
- It also matters "where you are" on your utility function

Motivation

"Deal or No Deal"

- 26 monetary values are "hidden" in briefcases
- A contestant picks a single briefcase to be their own (they do not know the value contained inside).
- They then have to select other briefcases to open; each time one is opened, its value is eliminated.
- Occasionally, a "banker" will make an offer of money to the contestant: they can accept the offer of money to leave the game, or they can say "no deal" and continue eliminating briefcases.
- If they eliminate all other values and don't accept an offer from the banker, they are left with the value contained in their original briefcase.

“Deal or No Deal” – How to Decide

Three important concepts we will discuss today:

“Expected Value”

- The remaining values have the same odds of being in your briefcase; as values are eliminated, the “expected value” of your briefcase changes.

“Certainty Equivalent”

- When the banker makes an offer, it is an offer to exchange a “sure bet” for a “risk”. What is the lowest offer you would accept?

“Risk Premium”

- If someone is risk-averse, they may accept less than their expected value as a certainty equivalent. The difference between the two is the “risk premium”.

"Deal or No Deal"

Deal or No Deal?

And the outcome…

Insurance

Question: Why do people and firms buy insurance?

The situation

- You own a beachfront restaurant on the US Atlantic (or Gulf) coast.
- Your beachfront business earns $200K per year and has $50K in the bank.
- Your business has a risk of a serious flood of 0.22% this year. The cost of rebuilding your business if there is a serious flood is $200K.
- A flood insurance policy that pays out $200K costs an annual premium of $500.

How should people/firms think about buying insurance?
Is it worth it?

We need to understand risk: there are multiple possible outcomes, and insurance changes our payoffs under some of them.

Uncertainty and Expected Value

Uncertainty

When an outcome is uncertain, we represent the potential outcomes by their payoffs, $\{x_1, x_2, \dots\}$, and their corresponding probabilities of being realized, $\{p_1, p_2, \dots\}$

Without Insurance

Without insurance, the uncertain outcomes you face are

$$\begin{array}{lclclcl} x_1 & = & \$50K & \quad & x_2 & = & \$250K \\ p_1 & = & 0.0022 & \quad & p_2 & = & 0.9978 \end{array}$$

With Insurance

With the $200K policy, you no longer face risk; flood or not, you have $249,500. That is, the $200K policy would **fully insure** you.

Definition: Expected Value

The **expected value** of an uncertain outcome is the sum

$$p_1 \cdot x_1 + p_2 \cdot x_2 + \cdots$$

For you, the expected values of your options are

- $249,500 if you buy the insurance
- $99.78\% \times \$250K + 0.22\% \times \$50K = \$249{,}560$ if you do not.

In other words, you lose $60 in expectation by buying the insurance.

Risk aversion

If you are risk-neutral, you will choose **not** to buy insurance.

- For comparison, private insurers in low-risk areas offer $200K flood coverage for closer to $750.
- Why would anyone buy this policy if it doesn't make sense for you to buy a $500 policy? Why are insurers in business?

Risk Aversion

- Many people would prefer a situation with a more certain outcome over a situation with a less certain outcome, even if the first has a lower expected value. These people are called **risk averse**.
- Indifference to risk is referred to as being **risk-neutral**, which some people and firms can also be **risk-seeking**, preferring uncertainty to certainty all else equal.

Since many people are risk-averse, they are willing to sacrifice some expected value in exchange for less variance in outcomes.

Risk aversion

How should we model this behavior?

Insight

The heart of risk aversion is the human tendency to worry more about the downside than the upside

Idea

What if we did an expected value calculation, but we "penalized" the upside?

Utility Function

Utility: have a function that translates payoffs so that its expected value reflects individual behavior.

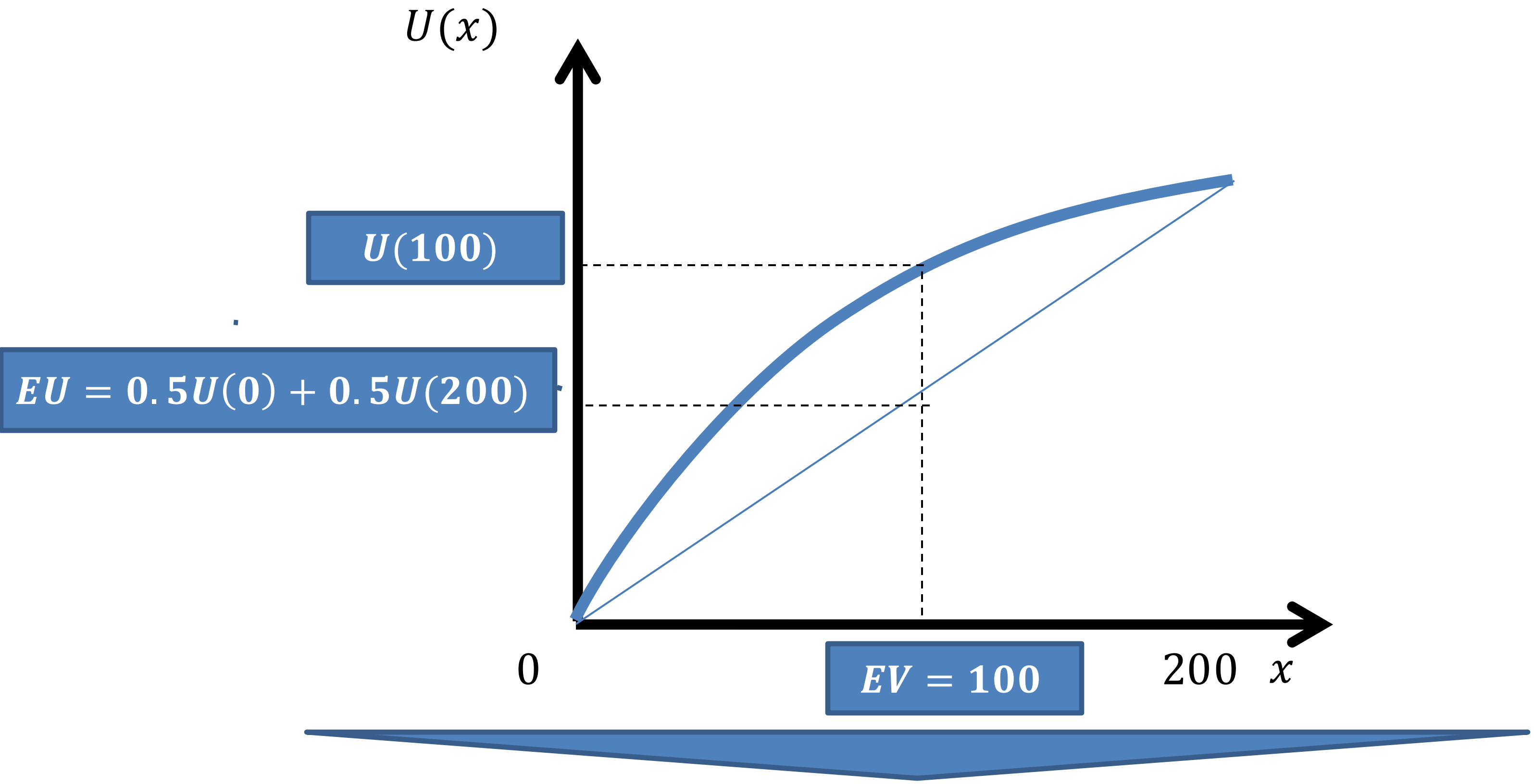
Expected Utility

Define a utility from an uncertain outcome x_1 as $u(x_1)$. Then, the expected utility is

$$p_1 \cdot u(x_1) + p_2 \cdot u(x_2) + \cdots$$

Let's see if we can come up with a utility function that will prefer $100K with certainty to a 50-50, $0-$200K uncertain outcome.

Expected Utility

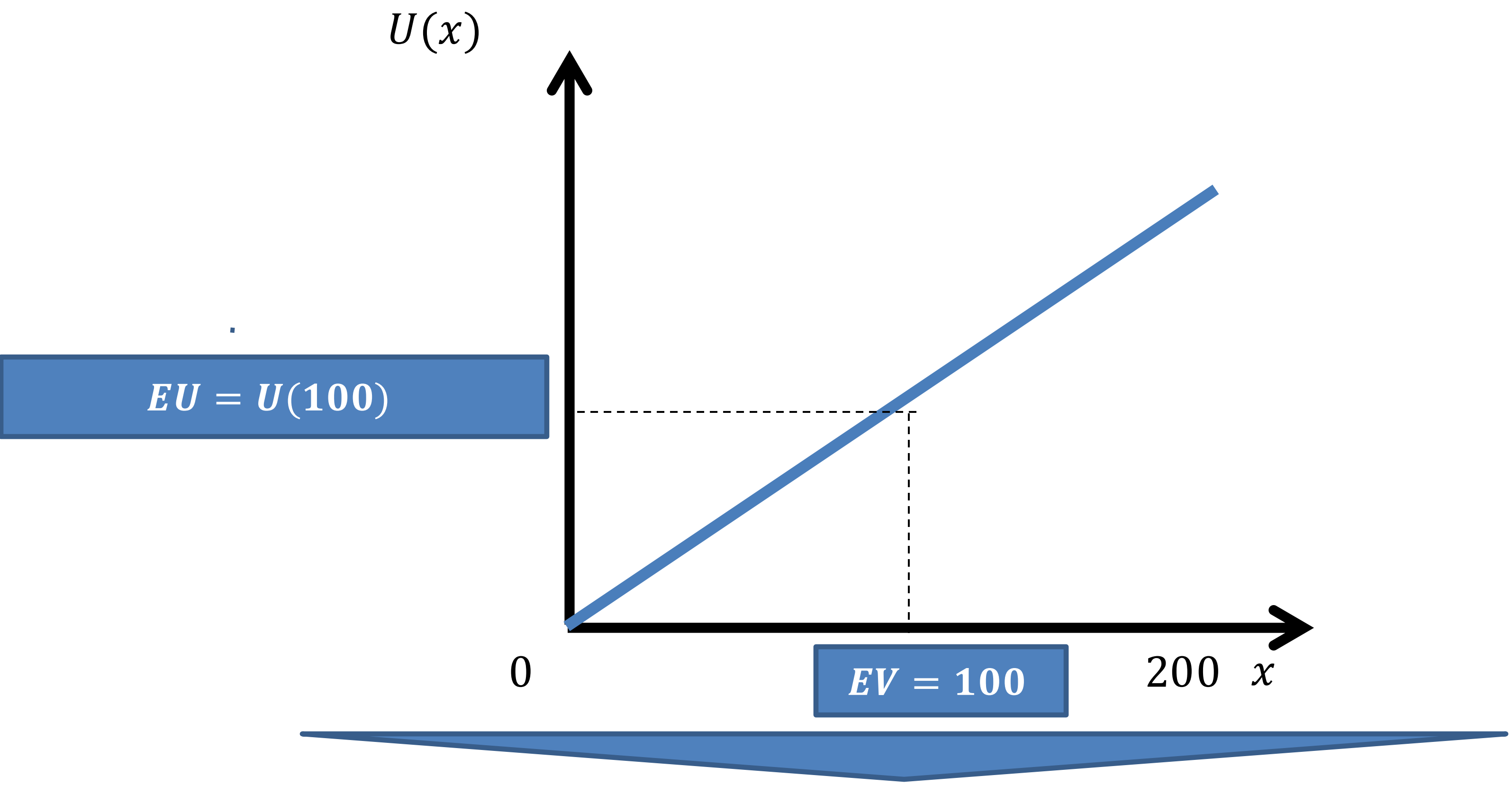


With a curved, concave utility function, we can get that

$$U(EV) > EU = 0.5U(0) + 0.5U(200)$$

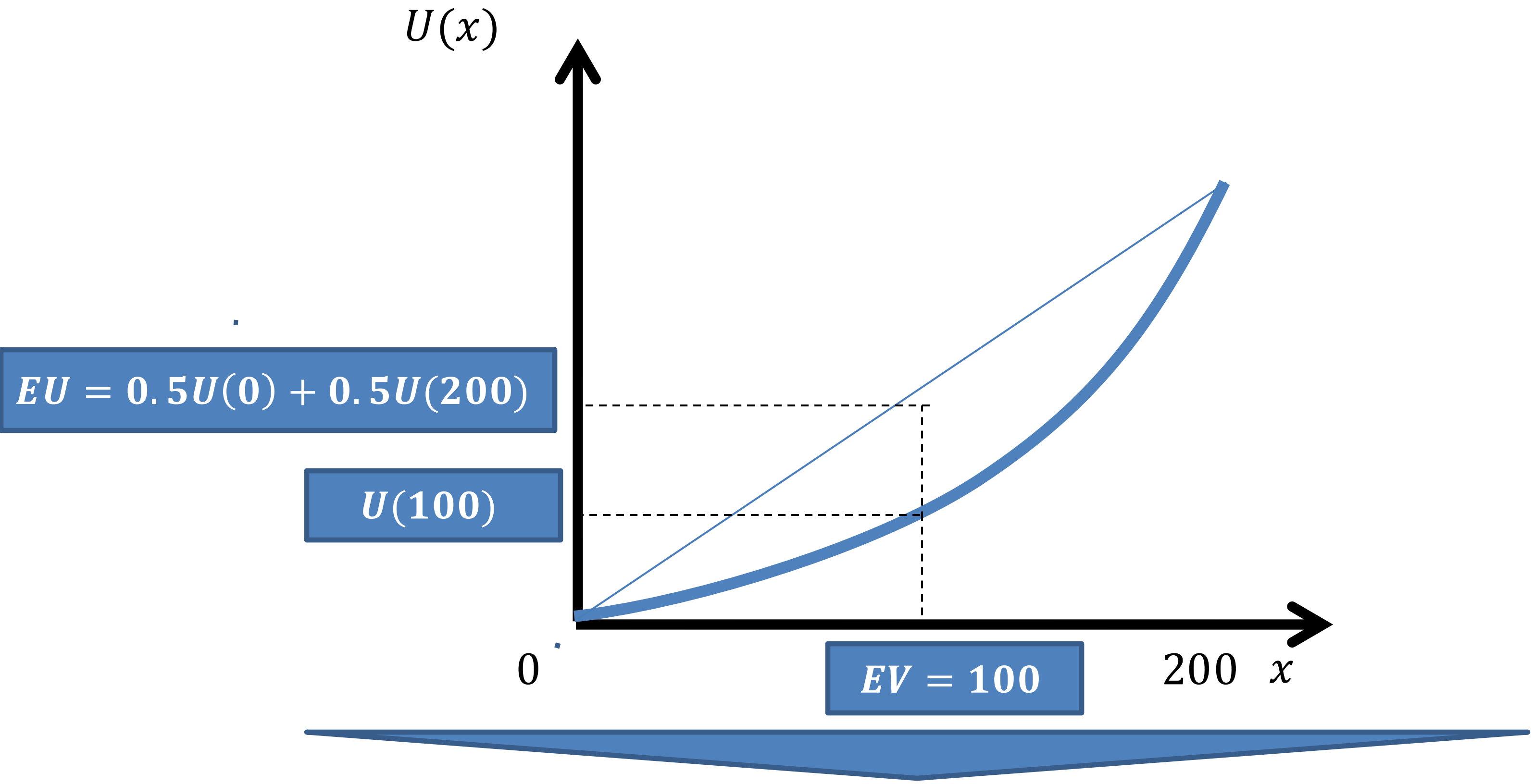
So, a person choosing based on expected utility would prefer the sure bet of 100 over the risk between 0 and 200. This person is **risk-averse.**

Risk neutrality



With a straight utility function, $U(EV) = EU = 0.5U(0) + 0.5U(200)$.
The person is indifferent between a risky bet and a sure outcome with the same expected value. We say this person is **risk neutral.**

Love for risk



With a convex utility function, $U(EV) < EU = 0.5U(0) + 0.5U(200)$.
This person is **risk-loving**: They prefer a risky bet over a sure outcome with the same expected value.

The form of a utility function

What curve can we choose to represent risk aversion?

Criteria for a Risk-Averse Utility Function

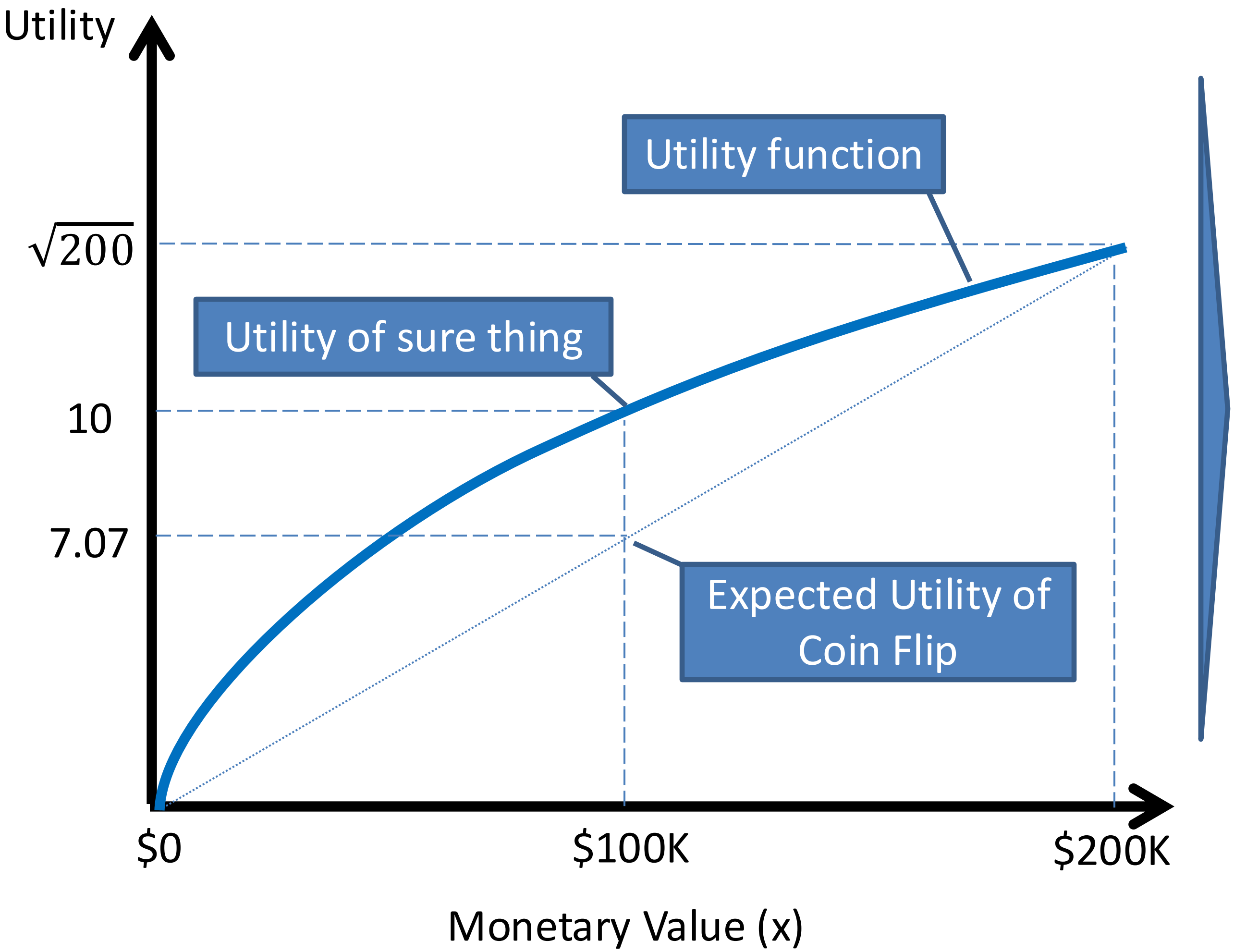
Increasing in x	A higher payoff is always better…
… but at a decreasing rate.	The first dollar of payoff is worth more than subsequent ones.

This will capture the idea that we are more sensitive to “downside” than “upside”.

Mathematically, we want the **marginal utility** to be positive, but decreasing.

$$MU(x) = u'(x) > 0$$
$$MU'(x) = u''(x) < 0$$

Example: $u(x) = \sqrt{x}$



Which of the following would you prefer:

- \$100K for sure, or
- \$0 and \$200K with equal probability.

The expected monetary value of both propositions is \$100K

The expected utility of the propositions is different:

- \$100K with certainty is worth $\sqrt{100} = 10$
- The "coin flip" is worth $\frac{1}{2} \cdot \sqrt{0} + \frac{1}{2} \cdot \sqrt{200} \approx 7.07$
- The sure thing has higher expected utility.

Buying Flood Insurance

The Situation

Let's return to our example from the beginning of class:

- 0.22% ($= p_d$) chance that there is a catastrophic flood that damages your business this year
- Outcome if there's a flood: \$50K
- Outcome if there isn't: \$250K
- Premium for \$200K of coverage: \$500

We will assume your utility function is $u(x) = \ln x$

…

What's $ln(x)$?

Math review:

- $\ln(x)$ is a natural logarithm
- Logarithms are an exponent at which a "base" is raised to yield a value.
- A natural logarithm has as its base the constant e.
- So, $\ln(5)$ is the exponent such that $e^{\ln(5)} = 5$.

It is a well-behaved concave function, so is often used in utility functions for risk aversion.

Now that we're caught up, let's figure out insurance.

Do We Buy the Insurance?

Our options are to decide between:

Full Insurance

$$\begin{aligned} EU &= \ln(\$250K - \$500) \\ &= 12.4272 \end{aligned}$$

No Insurance

$$\begin{aligned} EU &= p_d \cdot \ln(\$50K) + (1 - p_d) \cdot \ln(\$250K) \\ &= 12.4257 \end{aligned}$$

Yes, we will buy the insurance. How much is the insurance worth?

… 0.0015 “utils”.

We’re able to figure out a decision, but not *price* it.

How useful is this? Let’s figure out how to price it.

Today's Class

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Certainty Equivalent and Risk Premium

We need two concepts for thinking about the price of risk:

Certainty Equivalent

Amount you want to receive with certainty to be at least as well off as you are with the uncertain outcome.

Risk Premium

The difference between the expected value and the certainty equivalent of an uncertain outcome.

- A measure of how much you discount down from EV due to risk.

Suppose you were holding a lottery ticket that had a 1% chance of being worth \$1M, and a 99% chance of being worthless. (So, EV=\$10K).
What is the lowest price at which you would sell it to me?

This is your ***certainty equivalent***, denoted by c. For your utility function,

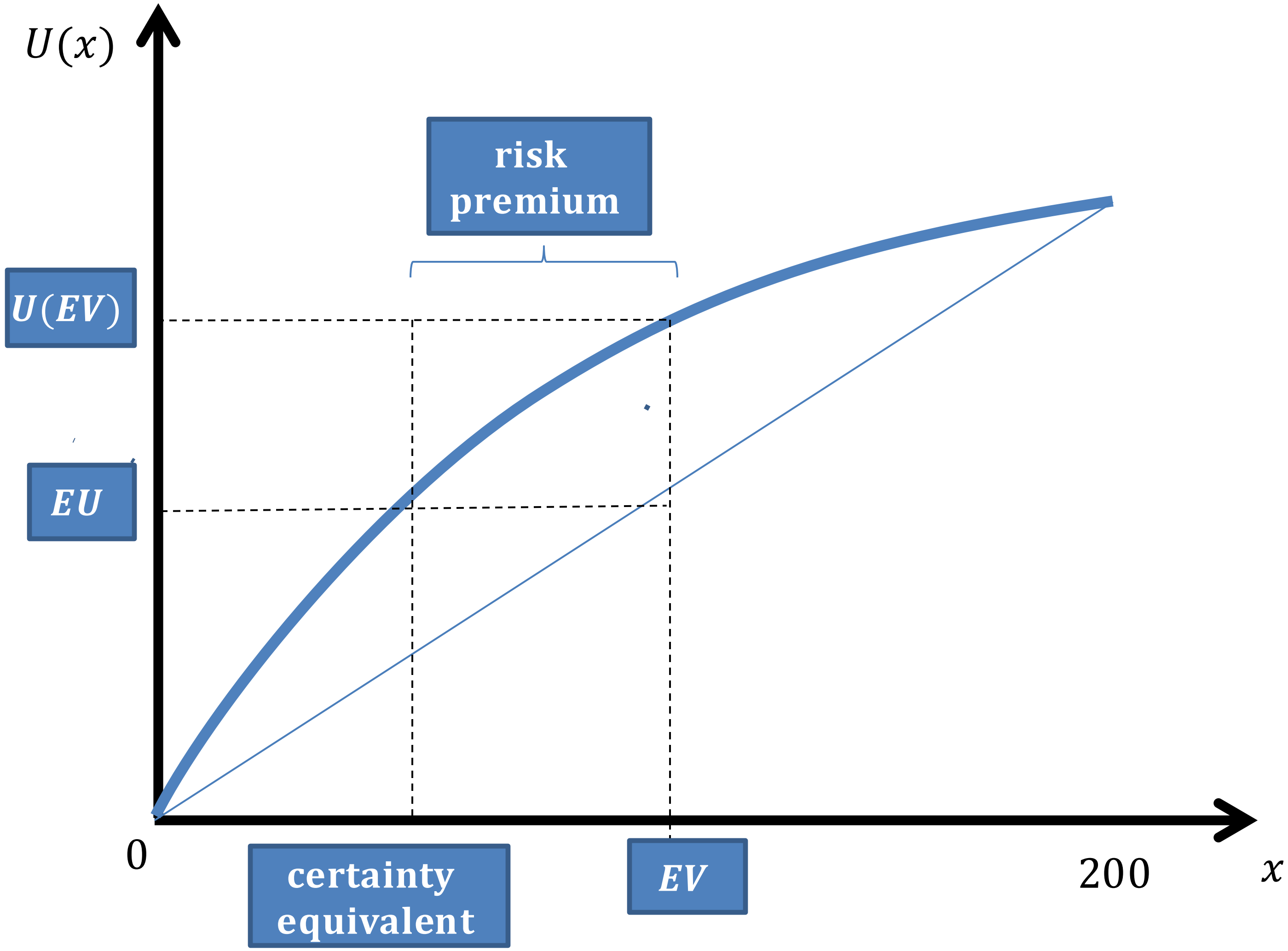
$$EU(risk) = U(certain\ c)$$

$$1\% \cdot u(\$1M) + 99\% \cdot u(\$0) = 100\% \cdot u(c)$$

Your ***risk premium*** is then:

$$r = EV - c = \$10{,}000 - \$c$$

Certainty Equivalent and Risk Premium



Willingness to Pay for Insurance

For the flood insurance, where $u(x) = \ln(x)$, we have:

The certainty equivalent of no insurance policy at all, c, is

$$\begin{aligned} EU(risk) &= EU(certain\ c) \\ p_d \cdot \ln \$50K + (1 - p_d) \cdot \ln \$250K &= \ln c \\ \ln[(\$50K)^{p_d} \cdot (\$250K)^{1-p_d}] &= \ln c \\ (\$50K)^{p_d} \cdot (\$250K)^{1-p_d} &= c \\ \$249{,}116 &= c \end{aligned}$$

Recall that the EV of the no insurance scenario is $249,560. That means that your risk premium is

$$r = EV - c = \$249{,}560 - \$249{,}116$$

i.e. you discount the EV of having no insurance down by $444 due to risk.

So, an insurance policy that guarantees me an outcome of $249,500 is a good deal!
In fact, I would be willing to pay more – up to $884 ($250,000-$249,116)

Insurance Concepts

	Concept	Our setting
Expected Value	The risk-neutral value of an uncertain outcome	99.78%×\$250K+0.22%×\$50 =\$249,560
Certainty Equivalent	The sure bet you would be willing to trade for the uncertain outcome.	$\ln(c)$ $= p_d \cdot \ln \$50K + (1 - p_d) \cdot \ln \$250K$ $c = \$249{,}116$
Risk Premium	The "cost" of the riskiness to you	\$444

People who are risk averse are willing to pay a risk premium to avoid uncertainty.

As you become more risk-averse, the risk premium is larger

Is Insurance a Good Thing?

Let's consider the two cases:

No Insurance Exists

You have a certainty equivalent of $249,116

Insurance: $200K for $500

You have a certainty equivalent of $249,500

- The payoff is $250,000 less $500 regardless.

Consumer surplus here goes up by…

- $384! Even though this insurance seemed like a bad deal from an expected value point of view, this seems like a good deal.
- The willingness-to-pay for full insurance is $250K-$c = $884.
  - Since this is full insurance, buying it guarantees you $250K-$p, where the premium is $p.
  - Here, they would pay up to $884 for this insurance.

What Affects Certainty Equivalent and Risk Premium?

Certainty equivalent and risk premium are affected by…

My Particular Utility Function

My utility function determines my level of risk aversion. More curvature means more risk-averse. For example:

1. $u(x) = x$: risk-neutral.
2. $u(x) = \ln(x)$: risk-averse (concave)
3. $u(x) = x^2$: risk-seeking (convex)

Where I am on my Utility Function

More curvature means more risk-averse; "savings" or an "outside option" affect where I am on my curve

My "savings" (outside option) and my level of risk aversion affect my certainty equivalent (and so my risk premium).

Example: Deal or No Deal

Assume your utility is ln(x). Consider two scenarios:

Available briefcases	EV	Certainty equivalent	Risk premium
200K, 400K	• 300K	• 283K	• 17K
100K, 500K	• 300K	• 224K	• 76K

Is your risk premium the same across the two scenarios?

Risk premium is higher in the second scenario because, in case of a bad draw, I am *twice as poor* as in the first.

Back to Our Show

Suppose we are left with the following values:

- $50, $100, $1,000, $75,000, and $1M.

"Expected Value"

- Each value has equal odds of being in my briefcase: 20%
- $EV = 20\% * (50 + 100 + 1K + 75K + 1M)$
- $215,230

"Certainty Equivalent"

- What is the smallest offer from the bank you would accept to quit the game?

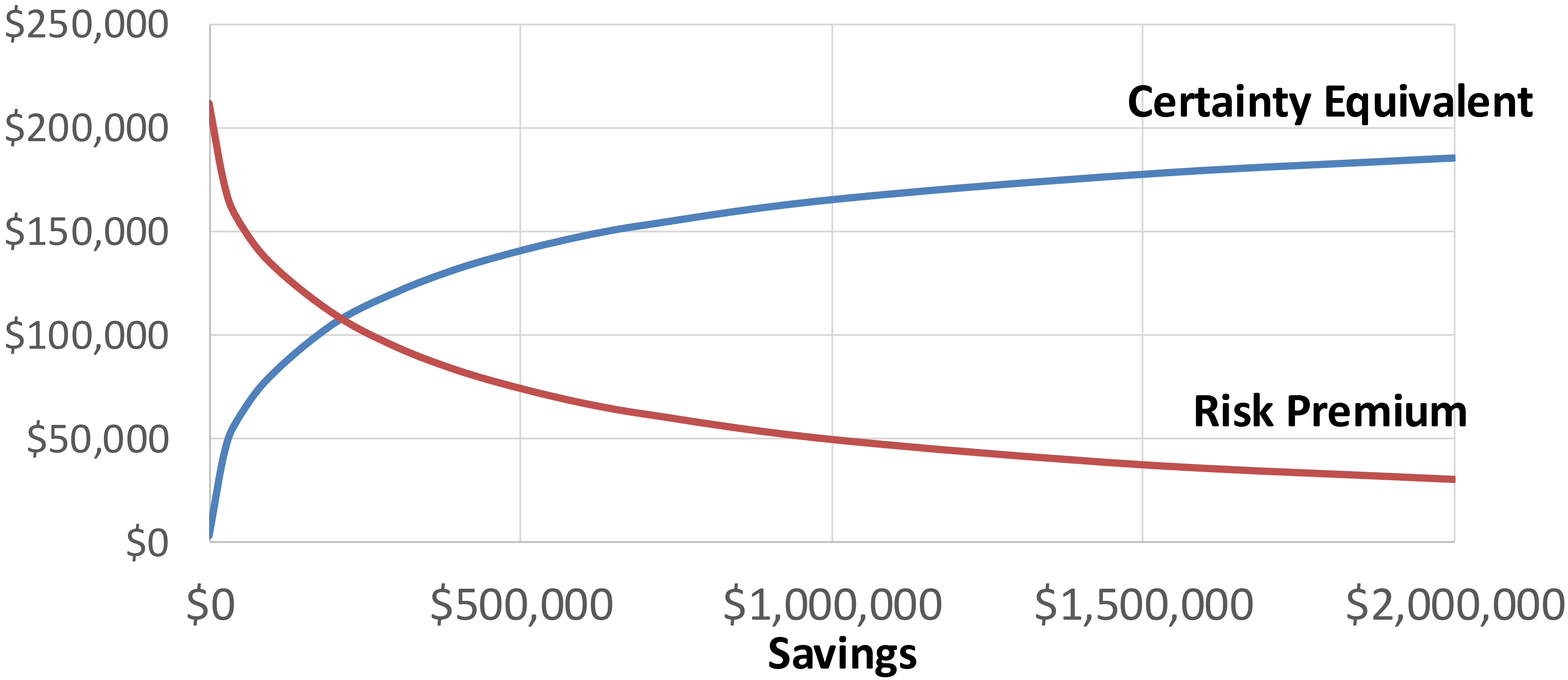
"Risk Premium"

- What is the difference between the EV and your certainty equivalent?

Question: what affects your certainty equivalent and risk premium?

Example: Deal or No Deal and Savings

Recall: Equal odds of \$50, \$100, \$1,000, \$75,000, and \$1M. EV=\$215,230
Suppose my utility function is $u(x) = \ln(x)$.



As I have more savings… I am less risk-averse (there is less downside). So, I am only willing to accept a higher Certainty Equivalent (or pay a lower Risk Premium)


BANK OFFER
$561,000
$ .01
$ 1
$ 5
$ 10
$ 25
$ 50
$ 75
$ 100
$ 200
$ 300
$ 400
$ 500
$ 1,000
$ 5,000
$ 10,000
$ 25,000
$ 50,000
$ 75,000
$ 100,000
$ 200,000
$ 1,000,000
$ 1,000,000
$ 1,000,000
$ 1,000,000
$1,000,000

Yale

The outcome…

Review and Today's Plan

Today

Making decisions based on expected value does not capture the desire to avoid risk
Risk-aversion can modeled with expected utility, where the utility function has positive, but decreasing, marginal utility.
To have an idea of how much better one prospect is than another, we must turn to certainty equivalent and the risk premium.

Coming up

Situations where parties do not have perfect information
Next: Adverse selection (hidden "type")
Then: Moral hazard (hidden action)